A periodic counterexample to the generated Taylor-polynomial zero-set conjecture

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Abstract

Conjecture 47 of Chung asserts a circular zero-set limit for a rationally generated polynomial sequence when its fixed quadratic factor has nonpositive discriminant. The admissible choice $a = b = c = 1$ refutes the conjecture: one infinite subsequence has all zeros on the unit circle, whereas a second has an additional zero at the origin. Consequently the upper and lower zero-set limits differ, and the limit required by the source does not exist.

Task 1: Resolve MathDB problem 358323

The problem, Conjecture 47 on printed page 119 of [1], concerns the polynomials defined by

$$\sum_{m\ge 0} P_m(z)t^m = \frac{1}{(at^2+bt+c)(1-tz)}, \qquad a,b,c \in \mathbb{R}\setminus\{0\}, \quad b^2-4ac \le 0. \tag{1}$$

If t_1 is a zero of at^2+bt+c of smallest modulus, the proposed conclusion is

$$\lim_{m\to\infty} Z(P_m) = \{z \in \mathbb{C} : |z| = 1/|t_1|\}.$$

Here $Z(P)$ is the set of complex zeros of P. Definitions 38–41 on printed pages 92–93 of the source specify the meaning of the limit. Writing $B(w,r) = \{z : |z-w| < r\}$, they are the usual set limits

$$w \in \limsup_m Z(P_m) \iff \text{every } B(w,r) \text{ meets } Z(P_m) \text{ for arbitrarily large } m,$$
$$w \in \liminf_m Z(P_m) \iff \text{every } B(w,r) \text{ meets } Z(P_m) \text{ for all sufficiently large } m.$$

The source defines $\lim_m Z(P_m)$ when these two sets are equal.

Theorem 1. *For $a = b = c = 1$ in (1), and every integer $r \ge 0$,*

$$P_{3r+1}(z) = (z-1)\sum_{j=0}^{r} z^{3j}, \tag{2}$$

$$P_{3r+2}(z) = z(z-1)\sum_{j=0}^{r} z^{3j}. \tag{3}$$

In particular,

$$0 \in \limsup_m Z(P_m) \setminus \liminf_m Z(P_m).$$

Thus Conjecture 47 is false even for positive a, b, c and strictly negative discriminant.

Proof. The parameters are nonzero and $b^2 - 4ac = -3 < 0$. The two quadratic zeros are $e^{2\pi i/3}$ and $e^{-2\pi i/3}$; both have modulus one. Thus the conjectured circle is $|z| = 1$.

Define coefficients c_k by

$$\frac{1}{1+t+t^2} = \frac{1-t}{1-t^3} = \sum_{j\geq 0}(t^{3j} - t^{3j+1}).$$

These identities hold as formal power series, so

$$c_{3j} = 1, \qquad c_{3j+1} = -1, \qquad c_{3j+2} = 0.$$

Multiplication by $(1-tz)^{-1}$ gives

$$P_m(z) = \sum_{k=0}^{m} c_k z^{m-k}. \tag{4}$$

For $m = 3r+1$, grouping the nonzero coefficients in pairs yields

$$P_{3r+1}(z) = \sum_{j=0}^{r}\left(z^{3r+1-3j} - z^{3r-3j}\right) = (z-1)\sum_{j=0}^{r} z^{3j}.$$

For $m = 3r+2$ the extra coefficient c_{3r+2} is zero, so (4) gives $P_{3r+2}(z) = zP_{3r+1}(z)$. This proves both factorization identities without division by a polynomial.

Every zero of $S_r(z) = \sum_{j=0}^{r} z^{3j}$ lies on the unit circle: if $S_r(z) = 0$, the polynomial identity

$$(z^3 - 1)S_r(z) = z^{3r+3} - 1$$

implies $z^{3r+3} = 1$ and hence $|z| = 1$. (For $r = 0$, $S_r = 1$ has no zeros.) Formula (2) therefore places every zero of P_{3r+1} on $|z| = 1$. Formula (3) places a zero of P_{3r+2} exactly at 0.

Every neighborhood of 0 consequently meets $Z(P_{3r+2})$ for every r. Thus 0 belongs to the upper limit. But the fixed ball $B(0, 1/2)$ is disjoint from $Z(P_{3r+1})$ for every r, so 0 does not belong to the lower limit. The required limit does not exist. Moreover, even if the conjectured limit were interpreted as the upper limit alone, its claimed circle would still omit the point 0. □

Scope. The theorem resolves the universal conjecture by an exact infinite-family counterexample. It makes no assertion about weak convergence of normalized zero-counting measures, which is a different question: a single exceptional zero has mass tending to zero in such a normalization.

Verification and status. The accompanying integer-arithmetic verifier constructs P_m independently from the cubic denominator in (1) and checks the coefficient and factorization identities. These finite checks supplement the proof for all r. A same-agent dependency audit and dated literature/status audit are saved alongside this file. MathDB was checked to be open with no posted claims on 11 September 2026; the searched primary sources disclosed no existing resolution. No external peer review is claimed.

Task 2: Classify the root-phase obstruction

For $b^2 - 4ac < 0$, express the quadratic zeros as $\rho e^{\pm i\theta}$. Determine which rational values of θ/π force the two zero-set limits to differ, and whether irrational values can have the same effect.

Task 3: Determine both limiting zero sets

Compute the full lower and upper zero-set limits for $a = b = c = 1$, including the subsequence $m \equiv 0 \pmod 3$; then extend the description to arbitrary nonzero real a, b, c with nonpositive discriminant.

Task 4: Establish a measure version of the limit

For the polynomials in (1), determine whether the normalized zero-counting measures converge weakly to uniform arclength on $|z| = 1/|t_1|$ throughout the nonpositive-discriminant range, with zeros counted with multiplicity.

Task 5: Quantify exceptional zeros

For each fixed admissible quadratic and $\varepsilon > 0$, bound the number of zeros lying outside the annulus $\big||z| - 1/|t_1|\big| < \varepsilon$ and determine the possible limits of these exceptional zeros along subsequences.

References

[1] J. Chung, *Zero Distribution of Generated Taylor Polynomials*, MSc project, California State University, Fresno, May 2022; arXiv:2305.05130v1, 9 May 2023. Definitions 38–41, pp. 92–93; Conjecture 47, p. 119.